

Generative Paper 2: Vacuum as the Minimal Condition of Definability

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Abstract

The concept of vacuum has undergone profound changes throughout the history of physics. In classical physics, vacuum was understood as empty space. In quantum field theory, it became the lowest-energy state of quantum fields, populated by fluctuations and virtual particles. Although these descriptions differ substantially, they share a common assumption: the vacuum is defined within an already existing physical universe. Space, time, geometry, and physical laws are all presupposed before the vacuum is introduced.

This paper explores a more fundamental question:

What remains if every one of these assumptions is removed?

Rather than treating the vacuum as a physical entity, we propose that it should be understood as the **minimal condition of definability**. At this level, the vacuum contains no geometry, no spacetime, no fields, no particles, no energy, and no information. It is not a physical background but the logical foundation that makes the emergence of physical structure possible.

We argue that such a vacuum requires only two primitive components: a **Domain**, which provides the possibility for definition, and a **Mapping**, which realizes the act of definition. Every additional concept—including coordinates, geometry, fields, and physical laws—already represents further structure and therefore cannot belong to the vacuum itself.

The objective of this paper is not to derive physical reality, but to establish the most minimal and internally consistent definition of vacuum upon which progressively richer mathematical and physical structures may later be constructed.

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1. Introduction

The meaning of the vacuum has evolved alongside the development of physics.

In classical mechanics, the vacuum was regarded simply as empty space—a region devoid of matter while still possessing the geometry of space and the flow of time. With the advent of quantum field theory, this picture changed dramatically. The vacuum came to be understood as the lowest-energy state of quantum fields, where fluctuations and virtual particles continuously arise as consequences of the underlying dynamics.

Despite their differences, these two conceptions share a common foundation. Both assume that the vacuum already exists within a physical universe. Space already exists. Time already exists. Geometry is already defined. Physical laws are already in operation.

For most areas of physics, these assumptions are entirely appropriate. Once the existence of spacetime and physical laws is accepted, the behavior of the vacuum becomes a well-defined scientific question. However, if the objective is to investigate the most fundamental layer of reality, these assumptions themselves require explanation.

This leads to a more primitive question:

What remains if every physical assumption is removed?

Suppose we remove space. There is no longer any notion of position or distance.

Suppose we remove time. There is no longer any ordering of events.

Suppose we remove particles, fields, energy, and information. There are no physical objects, no interactions, no measurable quantities, and no distinguishable states.

At this point, even the familiar concept of vacuum appears to disappear, since every traditional definition presupposes some form of physical structure.

Yet complete absence cannot serve as the foundation of a theory. If absolutely nothing existed—not even the possibility of definition—then no mathematical description, physical model, or logical statement could ever be formulated. A theory of fundamental reality therefore requires a condition that is simpler than any physical object, yet sufficient to make definition itself possible.

This observation motivates the central proposal of the present work.

The vacuum should not be identified with empty space, nor with the quantum vacuum described by modern field theory. Instead, it should be understood as the **minimal condition under which anything can become definable**. It is not a physical object but a pre-physical foundation. Its role is not to describe existing structures, but to provide the minimal logical environment from which such structures may eventually emerge.

The purpose of this paper is to investigate the consequences of adopting this viewpoint. Rather than asking what exists *within* the vacuum, we ask what the vacuum itself must minimally contain in order for any form of existence to become possible. The following sections show that this

requirement leads naturally to a remarkably simple structure consisting of only two primitive elements: a **Domain** and a **Mapping**.

2. The Need for a Foundational Vacuum

The previous section argued that every conventional definition of vacuum presupposes a physical framework. Whether the vacuum is described as empty space or as the ground state of quantum fields, it is always introduced after concepts such as space, time, geometry, and physical law have already been established.

If the goal is to investigate the deepest foundation of reality, however, this order must be reversed.

Rather than asking *what exists within the vacuum*, we must first ask **what the vacuum itself must be before any physical structure can exist**.

This shift changes the nature of the problem. The objective is no longer to characterize a physical state, but to identify the minimal condition required for the possibility of definition itself.

To accomplish this, we temporarily suspend every concept that already belongs to physics.

There is no space, and therefore no positions, distances, or geometry.

There is no time, and therefore no temporal ordering or evolution.

There are no particles, fields, or energy, and therefore no physical interactions.

There is no information, since information already presupposes distinguishable states capable of encoding differences.

Nothing remains that could properly be called a physical object.

The immediate temptation is to conclude that the vacuum is therefore simply "nothing." However, this conclusion is problematic. Absolute nothingness cannot serve as the foundation of a theory, because it provides no basis upon which anything could ever become distinguishable or describable. If even the possibility of definition is absent, then the concepts of existence and non-existence become equally meaningless.

A foundational vacuum must therefore satisfy a subtle but essential requirement.

It must be **minimal**, because every additional assumption introduces unnecessary structure.

At the same time, it must be **sufficient**, because complete absence cannot support the emergence of any definable reality.

The vacuum proposed in this work is intended to satisfy precisely these two requirements.

It is not an empty region of space, because space has not yet emerged.

It is not a physical medium, because no physical entities yet exist.

It is not an energetic state, because energy presupposes measurable quantities.

Instead, the vacuum is understood as the **minimal condition under which definition is possible**.

This interpretation deliberately separates the logical role of the vacuum from its physical interpretations. The vacuum is not the first physical object. Rather, it is the condition that makes the existence of physical objects conceivable in the first place.

This perspective naturally leads to the next question:

What is the smallest possible structure capable of supporting definability?

If the vacuum is truly minimal, then it cannot contain arbitrary mathematical objects or hidden physical ingredients. Every component must be indispensable, and every assumption must be justified.

The next section argues that only two primitive elements satisfy these requirements: a **Domain**, which provides the possibility for definition, and a **Mapping**, which realizes the act of definition. Together, they constitute the simplest structure capable of supporting a definable reality without introducing any unnecessary physical assumptions.

3. The Minimal Structure of Vacuum

If the vacuum is the minimal condition for definability, then its internal structure must also be minimal.

This requirement immediately constrains what the vacuum can contain. Every additional primitive concept increases the complexity of the foundation and therefore requires independent justification. A truly foundational theory should begin with no more structure than is strictly necessary.

The question is therefore not *what could* exist within the vacuum, but *what must* exist in order for definition itself to be possible.

We propose that only two primitive elements are required:

- **A Domain**
- **A Mapping**

Nothing else is assumed.

The **Domain** provides the possibility that distinctions can exist. It is the abstract substrate upon which definitions may be established. At this stage, the domain should not be interpreted as space, spacetime, a manifold, or any geometric object. Those concepts already possess structure that the vacuum is intended to precede.

Instead, the domain represents only the possibility that something can be defined.

The **Mapping** provides the complementary requirement. While the domain allows definitions to exist, the mapping performs the act of definition. It establishes relations within the domain and thereby distinguishes one potential state from another.

Without mapping, the domain remains completely undifferentiated. Although the possibility of definition exists, nothing is actually defined.

Conversely, a mapping cannot exist without a domain. Every mapping requires something upon which it acts. In the absence of a domain, the notion of mapping becomes meaningless.

The two elements therefore possess a fundamental interdependence.

A domain without mapping contains no distinctions.

A mapping without a domain has nothing to define.

Neither element is sufficient by itself.

Only together do they constitute the minimal structure capable of supporting definability.

This relationship can be represented schematically as

$$\text{Vacuum} = \text{Domain, Mapping}$$

This expression should not be interpreted as a physical decomposition of the vacuum into constituent parts. Rather, it identifies the two logical requirements that cannot be removed without destroying the possibility of definition itself.

The significance of this proposal lies in its minimality.

The domain introduces no geometry.

The mapping introduces no dynamics.

Together they introduce no particles, no fields, no energy, no information, and no physical laws.

They establish only the minimum logical framework within which definition can occur.

This interpretation also distinguishes the present framework from conventional mathematical constructions. In most mathematical theories, domains and mappings are introduced after an underlying structure has already been specified. The domain may be a set, a manifold, a vector space, or a topological space, while mappings operate within that predefined context.

Here, the situation is reversed.

The domain is not derived from a prior mathematical framework; it is itself one of the primitive assumptions. Likewise, the mapping is not introduced to describe physical processes or geometric transformations. Its sole role is to make definition possible at the most fundamental level.

The resulting vacuum is therefore intentionally featureless. It possesses neither physical properties nor mathematical richness beyond what is required for definability. Every familiar concept of physics and mathematics must be introduced only after this minimal structure has been established.

This principle of minimality has an important consequence. If any additional primitive element were included—such as coordinates, metrics, fields, particles, information, or physical constants—the vacuum would no longer represent the lowest possible layer of description. It would already contain structures whose own existence demands explanation.

The next section examines this point in detail by showing why concepts such as geometry, time, fields, energy, and information cannot belong to the vacuum itself, since each already presupposes a level of organization beyond the minimal structure established here.

4. Why Nothing Else Belongs to Vacuum

The previous section argued that the vacuum requires only two primitive elements: a **Domain**, which provides the possibility of definition, and a **Mapping**, which realizes the act of definition. The natural question is whether this structure is complete, or whether additional primitive concepts should also be included.

The central principle of the present framework is that the vacuum must remain **strictly minimal**. Every primitive concept introduced at this level becomes a foundational assumption. Consequently, no concept should be accepted unless it is indispensable for the possibility of definition itself.

This principle provides a simple criterion:

If a concept already presupposes organization, then it cannot belong to the vacuum.

Applying this criterion reveals that many of the concepts traditionally regarded as fundamental in physics are, in fact, already highly structured.

4.1 Space and Geometry

Space is often treated as the natural background of physical reality. However, the concept of space already assumes the existence of positions, separation, and geometric relationships.

Likewise, geometry presupposes notions such as distance, dimension, continuity, or topology.

These are not primitive ideas. They are organizational structures that distinguish one location from another and establish relationships between them.

If geometry were included in the vacuum, then the existence of geometric organization would already have been assumed rather than explained.

For this reason, neither space nor geometry can belong to the foundational vacuum.

4.2 Time

Time appears equally fundamental, yet it also contains implicit structure.

The existence of time already introduces an ordering of states. It distinguishes "before" from "after" and establishes the possibility of temporal evolution.

A vacuum intended to precede all physical organization cannot begin with such an ordering already in place.

Time must therefore be regarded as a structure that may emerge later, not as a primitive property of the vacuum.

4.3 Fields and Energy

Modern physics frequently describes the vacuum as a state of quantum fields possessing minimum energy.

However, both concepts already assume an extensive mathematical and physical framework.

A field requires values distributed over an underlying domain with established relationships.

Energy requires measurable states together with rules that distinguish one state from another.

Neither concept is meaningful in the absence of prior organization.

Consequently, fields and energy cannot be primitive constituents of the vacuum. They belong to a later stage of theoretical development, where physical structures have already become definable.

4.4 Information

Information is often regarded as one of the most fundamental concepts in contemporary science. Nevertheless, information itself presupposes distinction.

For information to exist, different states must already be identifiable. There must be recognizable differences capable of encoding meaning.

An entirely undifferentiated foundation cannot contain information, because information is itself a consequence of successful definition.

Within the present framework, information is therefore not primitive. It emerges only after definable structures have been established through the interaction of the Domain and the Mapping.

4.5 Physical Laws

The same reasoning applies to physical laws.

A law relates well-defined quantities according to well-defined rules. It presupposes that measurable objects, distinguishable states, and consistent mathematical relationships already exist.

The vacuum proposed here contains none of these.

Its purpose is not to specify how physical systems behave, but to provide the minimal condition under which physical systems can eventually become definable.

Physical laws therefore cannot be regarded as constituents of the vacuum. They represent higher levels of organization built upon a more primitive foundation.

4.6 The Principle of Minimality

The preceding examples illustrate a common pattern.

Every additional primitive assumption introduces new structure.

Every new structure introduces new concepts that themselves require explanation.

The search for a truly foundational description therefore demands that the primitive layer remain as simple as possible.

Within the framework developed in this paper, only the **Domain** and the **Mapping** satisfy this requirement. Together they provide the minimal logical conditions necessary for definition, while avoiding any assumptions about geometry, dynamics, physical entities, or mathematical organization.

The proposed vacuum is therefore intentionally sparse.

It is not empty because it lacks physical content.

It is minimal because it contains **only what is logically necessary for definability—and nothing more.**

This principle of minimality is the defining characteristic of the vacuum introduced in this work. It distinguishes the proposed framework from existing physical interpretations and provides the foundation upon which progressively richer mathematical and physical structures may later be constructed.

5. Conclusion and Outlook

This paper has reconsidered one of the oldest concepts in physics from a more fundamental perspective.

Rather than asking what the vacuum contains, we asked a prior question:

What is the minimal condition required for anything to become definable?

This shift changes the role of the vacuum completely. The vacuum is no longer interpreted as a physical region of empty space, nor as the lowest-energy state of quantum fields. Instead, it is understood as the minimal logical condition that makes the emergence of any definable structure possible.

Within this framework, the vacuum is intentionally stripped of every concept that already presupposes organization. It contains no space, no time, no geometry, no particles, no fields, no energy, no information, and no physical laws. Each of these concepts represents an additional layer of structure whose existence must itself be explained rather than assumed.

The analysis developed in this paper leads to a remarkably simple conclusion.

The minimal vacuum requires only two primitive elements:

- **A Domain**, providing the possibility for definition.
- **A Mapping**, realizing the act of definition.

Neither element is meaningful without the other. A domain without mapping remains completely undifferentiated, while a mapping without a domain has nothing upon which it can operate. Together, they constitute the smallest structure capable of supporting definability without introducing unnecessary assumptions.

The significance of this proposal lies not in adding a new physical entity to existing theories, but in identifying a more primitive level beneath them. The vacuum introduced here is not intended to replace the classical or quantum descriptions of vacuum. Those remain valid within the physical frameworks in which they are defined. Instead, the present work addresses a different question altogether: **what must exist before any physical framework can exist?**

Viewed from this perspective, the classical vacuum and the quantum vacuum become higher-level realizations of a more fundamental concept. They are physical manifestations that already assume geometry, spacetime, and physical law, whereas the foundational vacuum proposed here precedes all of these assumptions.

This paper therefore establishes the second foundational layer of the Generative Architecture. The first paper identified the **recursively self-generating function** as the minimal primitive from which recursive generation can begin. The present work identifies the **vacuum** as the minimal condition within which such generation can become definable.

Together, these two papers establish the conceptual foundation of the framework:

- **Paper 1:** *What is the minimal primitive?* → A recursively self-generating function.
- **Paper 2:** *What is the minimal condition for definability?* → A vacuum consisting only of a Domain and a Mapping.

The subsequent papers will build upon these foundations by investigating how progressively richer organizational structures emerge from this minimal setting. Rather than introducing physical concepts as primitive assumptions, they will examine how notions such as organization, interaction, geometry, and ultimately physical law can arise through successive stages of generative development.

In this way, the present work completes the definition of the foundational vacuum while preparing the conceptual ground for the emergence of increasingly structured descriptions of reality.

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This paper represents one step within the ongoing development of the **Generative Architecture** framework. Subsequent papers will progressively extend the concepts introduced here into a unified theoretical structure.